Course Information Sheet

Electrical Circuits (ET1213)

Document No. CRS-0050 · Academic Year 2024/2025 · Semester 2

Course Details

Field	Details
Course ID	12
Course Name	Electrical Circuits
Course Code	ET1213
Credit Value	3.00
Department	Engineering Technology (Department ID: 2)

Course Description

This course presents the fundamental principles of electrical circuit analysis for engineering technology students. It covers Ohm's law, Kirchhoff's laws, network theorems, and methods for analysing direct-current and alternating-current circuits, including phasor techniques. Resonance, power, and energy in circuits are examined, and basic measurement instruments are introduced in the laboratory. Safety practice when working with electrical systems is emphasized throughout the course.

Learning Outcomes

On successful completion of this course, students will be able to:

- Apply Ohm's law and Kirchhoff's laws to analyse electrical networks.
- Use network theorems to simplify and solve direct-current circuits.
- Analyse alternating-current circuits using phasor methods, including resonance and power.
- Perform circuit measurements safely with standard laboratory instruments.

Assessment and Grading

Component	Weighting	Description
Assignments and Quizzes	20%	Graded coursework issued and completed during the semester
Mid-Semester Test	10%	Written test covering the first half of the course content
Final Examination	70%	End-of-semester written examination

Note: this is a sample document generated for project testing purposes. Assessment weights and descriptive text are illustrative.